User-Constructed Information Architecture as a Persistence Layer for GenUI

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Generative UI produces a new interface for every interaction, but the information tasks these interfaces support–comparing options, planning trips, researching decisions–unfold across multiple queries and iterations. During these tasks, users implicitly construct their own organizational structure: categories, priorities, terms, and role relationships. In current GenUI systems, this structure is rarely captured, and typically lost between generations. We propose adapting the design practice of Information Architecture (IA) as a persistence layer that captures user-constructed structures and applies it across generations, transforming ephemeral interfaces into accumulative ones. We identify four elements–Partition, Hierarchy, Order, and Vocabulary–adapting traditional IA as a shared layer between users and AI as a dimension of persistence. We discuss implications for design principles and how this layer could generalize across workflows.

CCS Concepts: • **Human-centered computing → Human computer interaction (HCI)**.

Additional Key Words and Phrases: Generative User Interfaces, Information Architecture

1 The Persistence Problem

Generative UI (GenUI) produces a new interface for every interaction, enabling contextual and personalized responses. However, this ephemerality means that structure built during one interaction is lost in the next. While some systems persist conversation history [5] or task context [2] for GenUI, user-constructed structure–the categories users create, the priorities they develop, the vocabulary they define, the connections they draw across sources–is typically lost.

Yet information tasks–comparing apartments, planning trips, researching decisions–unfold over multiple queries and sessions, across different sources. Sensemaking research shows that users develop rich structures during extended tasks [9, 10]: schemas for evaluating options [8], personal terminology that evolves with understanding [11], and organizational structures that reflect task-specific needs [3, 13]. Current GenUI systems do not capture or persist these structures. We argue that GenUI needs a persistence layer that can be shared between the user and AI for user-constructed structures: the categories, priorities, terms, and connections users implicitly build. This layer should capture structure as it emerges, apply it across generations, and respect that different structures need different lifespans.

2 Information Architecture as a Persistence Layer

Traditional information architecture (IA), as articulated by Rosenfeld and Morville [12], concerns how information is organized, labeled, and connected within designed systems through organization schemes, labeling systems, and navigation structures. This architecture is typically author-designed and users inhabit structures created for them [6].

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During information tasks, however, users construct their own structures. They develop categories that reflect their goals, not source organization [3]. They articulate criteria and priorities through iterative comparison [4]. They build personal vocabulary whose meaning shifts with context. They connect fragments across sources that no single system links [7]. This user-constructed IA is implicit, distributed across mental effort and scattered notes, and lost when the task ends or the tool changes.

We reframe IA for the GenUI context. Rather than focusing on what structures authors design, we ask: what structures do users construct, and what should GenUI persist? We identify four elements.

- **Partition** captures user-created categories and groupings that emerge from task framing rather than source structure–adapting traditional *organization schemes* from author-designed to user-constructed. Users build categories like "Day 1 vs. Day 2" or "Consider vs. Ruled Out." In UI, this shapes section headers, groupings, and layout divisions.
- **Hierarchy** captures role-differentiated relationships, where an item's position signals what it is relative to its neighbors–extending *traditional hierarchy* into user-constructed semantic role relationships. Users place evidence clips under a specific listing, or nest criteria under broader categories. In UI, this shapes containment structures, nested components, and parent-child layouts.
- **Order** captures evaluative ranking of criteria, reflecting what dimensions matter and in what priority–extending *hierarchy* from taxonomic containment to evaluative prioritization. Users rank criteria like "walkability matters most, then cost." In UI, this shapes sort ranking, column arrangement, and what content is emphasized.
- **Vocabulary** captures personal vocabulary with contextual meaning, where terms are task-specific and user-determined–adapting *labeling and controlled vocabulary* from author-prescribed to user-constructed. "Affordable" or "nearby" carry meanings that shift between tasks. In UI, this shapes labels, badges, and filter criteria.

A GenUI system aware of this IA could generate outputs sorted by user priorities, labeled with user vocabulary, grouped by user categories, and structured by user-defined role relationships.

3 Implications for Practice

If GenUI systems adopt a persistence layer for user-constructed IA, several implications follow for HCI practice.

Designing for user-constructed IA. Systems that persist user-constructed IA must address several challenges. They need IA schemas that define what dimensions users can structure. Traditional IA offers guidelines for author-designed systems–principles like progressive disclosure, clear labeling, and consistent structure [1]. Which of these transfer to user-constructed IA? Some may apply directly: users still benefit from clear defaults and consistent vocabulary. Others may need adaptation: progressive disclosure assumes author control over what users see, but user-constructed IA shifts that control. New principles may also emerge–how should systems balance inference against explicit control? Developing design guidelines for user-constructed IA in GenUI remains an open area.

IA as a generalizable layer. The persistence layer we propose is not specific to one workflow or modality. Users might construct IA through chat, direct manipulation, or interaction with existing pages. This IA could then shape varied outputs beyond UI generation: a chat response structured by user priorities, a page overlay reorganizing content, or a generated document reflecting user vocabulary. How this layer integrates with existing LLM architectures and context managers–as structured context, plugin schemas, or portable artifacts–remains an open question.

In summary, we propose user-constructed IA as a bridge between GenUI's ephemeral nature and the extended temporality of real information tasks.

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